

Assignment 1 Submission; due Thursday, September 24, 2026

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- [1] Let $E \subseteq \mathbb{R}$ be nonempty. We say that $t \in \mathbb{R}$ is a *lower bound* of E if $t \leq x$ for all $x \in E$. We say that E is *bounded below* if there exists a lower bound of E . We say that $t \in \mathbb{R}$ is a *greatest lower bound*, or *infimum*, of E if it is a lower bound of E and if $t \geq s$ for any lower bound s of E . If E has an infimum, it is denoted $\inf E$.

- [a] Let E be a nonempty subset of $\mathbb{R}$ which is bounded below. Find a way to use the completeness axiom to prove that $\inf E$ exists. *Hint:* Consider the subset $-E = \{-x : x \in E\}$.

Answer:

Proof. Since E is nonempty and bounded below, there exists $\alpha \in \mathbb{R}$ such that

$$\forall x \in E : \alpha \leq x$$

Consequently,

$$\forall x \in E : -\alpha \geq -x$$

.

Define

$$-E = \{-x : x \in E\}.$$

Then, $-\alpha$ is an upper bound of $-E$.

By the completeness axiom, $\sup(-E)$ exists. Let

$$\beta = \sup(-E).$$

It follows that $\forall x \in -E : \beta \geq x$. And therefore, $\forall x \in E : -\beta \leq x$. So $-\beta$ is a lower bound of E .

Next, take any other lower bound of E , call it s . From the same logic as above, it follows that $-s$ is an upper bound of $-E$. Since β is a supremum of $-E$, it follows that $\beta \leq -s$.

So, $-\beta \geq s$. Therefore, since $-\beta$ is a lower bound of E and it is at least as large as any other lower bound, it follows that $-\beta = \inf E$ □

- [b] Let $\emptyset \neq A \subseteq B \subseteq \mathbb{R}$. Suppose B is bounded below (so $\inf B$ exists). Prove that A is bounded below (so $\inf A$ exists) and that $\inf A \geq \inf B$.

Answer:

Proof. If B is bounded below, then $\exists \alpha : \forall x \in B : \alpha \leq x$.

Since $A \subseteq B$, we know that $\forall y \in A : y \in B$.

Consequently,

$$\forall y \in A : \alpha \leq y$$

Therefore, α is also a lower bound of A . And from 1[a], we can conclude that $\inf A$ also exists.

Now, since A is a subset of B and $\inf B$ is a lower bound of B , then $\inf B$ must also be a lower bound of A . Since $\inf A$ must be greater than or equal to any other lower bounds of A , we have that

$$\inf A \geq \inf B$$

□

- [c] Let A, B be nonempty subsets of $\mathbb{R}$, both bounded below (so both $\inf A$ and $\inf B$ exist). Define

$$A + B = \{a + b : a \in A, b \in B\}.$$

Prove that $A + B$ is bounded below and that $\inf(A + B) = \inf A + \inf B$.

Answer:

Proof.

$$\begin{aligned} \forall a \in A : \inf A &\leq a \\ \forall b \in B : \inf B &\leq b \\ \therefore \forall a \in A, b \in B : \inf A + \inf B &\leq a + b \\ \therefore \forall x \in A + B : \inf A + \inf B &\leq x \end{aligned}$$

So $\inf A + \inf B$ is a lower bound of $A + B$. Consequently, $\inf(A + B)$ exists.

Now, we know that

$$\begin{aligned} \forall \varepsilon > 0 : \exists \alpha \in A : \inf A + \frac{\varepsilon}{2} &> \alpha \\ \forall \varepsilon > 0 : \exists \beta \in B : \inf B + \frac{\varepsilon}{2} &> \beta \end{aligned}$$

Therefore,

$$\forall \varepsilon > 0 : \exists \alpha \in A, \beta \in B : \inf A + \inf B + \varepsilon > \alpha + \beta$$

This means that no number greater than $\inf A + \inf B$ can be a lower bound of $A + B$. Therefore,

$$\inf(A + B) = \inf A + \inf B$$

□

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- [2] Let x, y be two real numbers such that $y - x > 1$. (In particular, $x < y$.)

- [a] If x is positive, prove that there exists an integer $k \in \mathbb{Z}$ such that $x < k < y$.

Answer:

Proof. By the Archimedean property, the set $E = \{n \in \mathbb{N} : n > x\} \subset \mathbb{N} \subset \mathbb{Z}$ is not empty.

x is a lower bound of E , so by the completeness axiom, $\inf E$ exists. And since $E \subset \mathbb{Z}$, $\inf E \in E$. Let $\inf E = k$.

Then $x < k$. Now, assume for the sake of contradiction that $x < k - 1$. Since $k - 1 \in \mathbb{Z}$, then $k - 1 \in E$. But $k - 1 < k$ and $k = \inf E$, which is a contradiction.

Therefore, $k - 1 \leq x$. $k \leq x + 1 < y$.

So, $x < k < y$. □

- [b] If x is negative and y is positive, prove that there exists an integer $k \in \mathbb{Z}$ such that $x < k < y$.

Answer:

Proof. Take $k = 0 \in \mathbb{Z}$. □

- [c] If y is negative, prove that there exists an integer $k \in \mathbb{Z}$ such that $x < k < y$.

Answer:

Proof. Since $y < 0$, we have $-y > 0$. Moreover,

$$(-x) - (-y) = y - x > 1$$

So applying 2[a] to $-x$ & $-y$, we get that $\exists n \in \mathbb{Z} : -y < n < -x$.

$$x < -n < y$$

Taking $k = -n \in \mathbb{Z}$ proves the result. □

- [3] The parts of this problem are *not related*.

- [a] Let $x \in \mathbb{R}$. Consider the set $E = \{q \in \mathbb{Q} : q < x\} \subset \mathbb{Q} \subset \mathbb{R}$. Show that E is nonempty and bounded above, so $\sup E$ exists. Show also that $\sup E = x$.

Answer:

Proof. Applying the Archimedean property to $-x$ we get that $\exists n \in \mathbb{N} : n > -x$. So, $x > -n$. Therefore, $-n \in E$, so E is non-empty.

Also, since x is strictly larger than any element of E , x is a lower bound of E .

Since E is bounded above and non-empty, $\sup E$ exists.

Let $\sup E = \alpha$. We know that $\alpha \leq x$. Now, assume for the sake of contradiction, that $\alpha < x$.

One of the propositions proved in class showed that between any two distinct real numbers, there exists a rational number. That is:

$$\exists q \in \mathbb{Q} : \alpha < q < x$$

But if q is rational and smaller than x , then it should be a part of E . But this contradicts α being an infimum of E . Therefore, our assumption must be false and

$$\sup E = x$$

□

- [b] Let $x, y \in \mathbb{R}$ with $x < y$. Prove that there exists an *irrational* (not rational) number z such that $x < z < y$. *Hint:* You may use the fact that $\sqrt{2}$ is irrational.

Answer:

Proof. We will take the interval (x, y) and shift it by $\sqrt{2}$.

$$x - \sqrt{2} < y - \sqrt{2}$$

By the proposition shown in class, we know that

$$\exists q \in \mathbb{Q} : x - \sqrt{2} < q < y - \sqrt{2}$$

Therefore, $x < q + \sqrt{2} < y$. Since we know that $\sqrt{2}$ is irrational, then $z = q + \sqrt{2}$ is also irrational. □

- [4] Let $C > 0$ be a fixed positive number.

- [a] Let $x \geq 0$, and suppose that for any $\varepsilon > 0$, we have $x < C\varepsilon$. Show that for any $\eta > 0$, we have $x < \eta$.

Answer:

Proof. Let $\eta > 0$ and choose

$$\varepsilon = \frac{\eta}{C}$$

Then we know that $x < C\varepsilon = C \frac{\eta}{C}$. Therefore, $x < \eta$ □

- [b] In the situation of part [a], show that we must have $x = 0$. *Hint:* Find a way to use the Archimedean property.

Answer:

Proof. Suppose for the sake of contradiction that $x > 0$.

By the Archimedean property, we get that

$$\exists n \in \mathbb{N} : n > \frac{1}{x}$$

Therefore, $x > \frac{1}{n}$. But we can take $\eta = \frac{1}{n} > 0$ in 4[a] and get that $x < \frac{1}{n}$. This is a contradiction, so our assumption must be wrong. Therefore, $x = 0$. □

- [5] In this problem you will give an alternative proof of the Cauchy–Schwarz inequality. The inequality says that

$$-\|x\|\|y\| \leq \langle x, y \rangle \leq \|x\|\|y\|.$$

Both inequalities are trivially seen to be equalities if either $x = 0$ or $y = 0$, so we suppose that both vectors are nonzero.

By consider the norm of the vector

$$z = \frac{x}{\|x\|} \pm \frac{y}{\|y\|},$$

show that the first inequality becomes an equality if and only if $y = tx$ for some $t < 0$, and that the second inequality becomes an equality if and only if $y = tx$ for some $t > 0$.

Answer:

Proof. Let $u = \frac{x}{\|x\|}$ and $v = \frac{y}{\|y\|}$.

We will first prove

$$y = tx \text{ for some } t > 0 \iff \langle x, y \rangle = \|x\|\|y\|$$

First, suppose $y = tx$ for some $t > 0$. We have to prove that $\langle x, y \rangle = \|x\|\|y\|$.

$$\langle x, y \rangle = \langle x, tx \rangle = t \langle x, x \rangle = t\|x\|^2$$

$$\|y\| = \|tx\| = t\|x\|$$

$$\therefore \langle x, y \rangle = \|x\|\|y\|$$

Secondly, suppose $\langle x, y \rangle = \|x\|\|y\|$. We have to prove that $y = tx$ for some $t > 0$.

Let us now look at $\langle u, v \rangle$

$$\langle u, v \rangle = \left\langle \frac{x}{\|x\|}, \frac{y}{\|y\|} \right\rangle = \frac{\langle x, y \rangle}{\|x\|\|y\|} = 1$$

Let $z = u - v$.

$$\|z\|^2 = \langle z, z \rangle = \langle u - v, u - v \rangle = \langle u, u \rangle - 2\langle u, v \rangle + \langle v, v \rangle$$

$$\|z\|^2 = \|u\|^2 + \|v\|^2 - 2$$

$$\|u\| = \left\| \frac{x}{\|x\|} \right\| = \frac{\|x\|}{\|x\|} = 1$$

$$\|v\| = \left\| \frac{y}{\|y\|} \right\| = \frac{\|y\|}{\|y\|} = 1$$

$$\|z\|^2 = 2 - 2 = 0$$

$$\|z\| = 0$$

$$z = u - v = 0$$

$$u = v$$

$$\frac{x}{\|x\|} = \frac{y}{\|y\|}$$

$$y = x \frac{\|y\|}{\|x\|}$$

$$\therefore y = tx$$

$$\|y\| > 0, \quad \|x\| > 0$$

$$\text{For } t = \frac{\|y\|}{\|x\|} > 0$$

This concludes the proof that $y = tx$ for some $t > 0 \iff \langle x, y \rangle = \|x\|\|y\|$

Now we have to prove

$$y = tx \text{ for some } t < 0 \iff \langle x, y \rangle = -\|x\|\|y\|$$

Consider the vector $x' = -x$ and apply the proof above to show that

$$y = t'x' \text{ for some } t' > 0 \iff \langle x', y \rangle = \|x'\|\|y\|$$

Now,

$$y = t'(-x) \text{ for some } t' > 0 \iff \langle (-x), y \rangle = \|(-x)\|\|y\|$$

$$y = (-t')x \text{ for some } t' > 0 \iff -\langle x, y \rangle = \|x\|\|y\| \quad \|x\| = \|-x\|$$

$$y = (-t')x \text{ for some } t' > 0 \iff \langle x, y \rangle = -\|x\|\|y\|$$

Take $t = -t' < 0$

$$y = tx \text{ for some } t < 0 \iff \langle x, y \rangle = -\|x\|\|y\|$$

□

[6] Let

$$E = \overline{B_r(x_0)} = \{x \in \mathbb{R}^n : \|x - x_0\| \leq r\}$$

be a closed ball. Prove that E is *not open*.

Answer:

[7] Let $F = \{x_1, \dots, x_N\}$ be a finite subset of $\mathbb{R}^n$. Prove that F is *closed*.

Answer:

[8] Let $a \in \mathbb{R}^n \setminus \{0\}$, and $c \in \mathbb{R}$. Consider the subset

$$E = \{x \in \mathbb{R}^n : \langle x, a \rangle < c\}.$$

Prove that E is open. *Hint:* Observe that

$$|\langle y, a \rangle - \langle z, a \rangle| = |\langle y - z, a \rangle| \leq \|y - z\|\|a\|$$

by Cauchy–Schwarz.

Answer:

[9] COMING SOON

Answer:

[10] COMING SOON

Answer: